

Clustering and Classification:

Netflix Content Analysis

- Kshitij Thorat

Executive Summary

This analysis employs machine learning techniques to classify Netflix content types (Movies vs. TV Shows) and discover underlying patterns through clustering. The Random Forest classifier achieves **perfect accuracy (100%)** on the test set, while K-Means clustering reveals three distinct content groupings based on temporal and duration characteristics.

1. Data Preprocessing & Feature Engineering

Data Loading & Cleaning

- Dataset loaded from Databricks Volumes using Spark's distributed computing framework
- Null value handling: Rows with missing values in critical columns (type, rating, release_year, duration_int, added_year) are removed to ensure data quality
- This ensures model training occurs on complete, reliable records

```
StringIndexer(inputCol="type", outputCol="type_idx")
```

```
StringIndexer(inputCol="rating", outputCol="rating_idx")
```

Purpose: Converts categorical variables into numerical representations required by ML algorithms

- type: Movie/TV Show → 0/1
- rating: G, PG, R, etc. → numerical indices
- handleInvalid="skip" prevents errors from unseen categories

Key Insights

The feature set captures multiple dimensions of content:

- **Temporal:** release_year, added_year - tracks content evolution
- **Duration:** duration_int - distinguishes short-form vs. long-form content
- **Categorical:** type_idx, rating_idx - content classification and audience targeting

2. Classification Analysis: Random Forest Model

Model Architecture

- Algorithm: Random Forest Classifier
- Number of trees: 100 (ensemble learning)
- Train-test split: 80-20 with seed=42 for reproducibility

Why Random Forest?

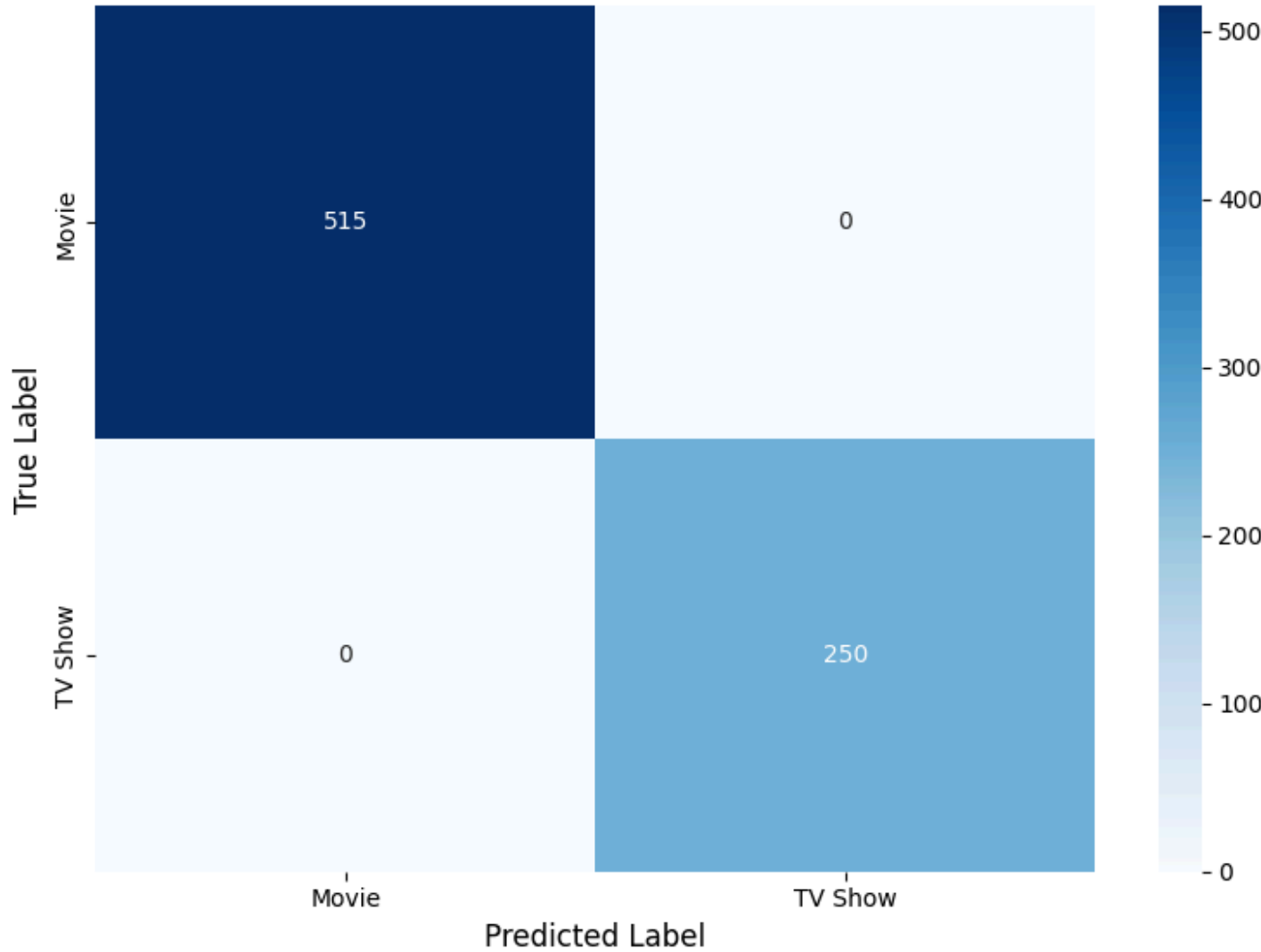
- 1. **Ensemble Learning:** Aggregates predictions from 100 decision trees, reducing overfiance
- 2. **Feature Importance:** Inherently provides interpretability through feature ranking
- 3. **Robustness:** Handles non-linear relationships and feature interactions
- 4. **No Feature Scaling Required:** Tree-based models are invariant to monotonic transformations

Classification Report Analysis

Metric	Movie	TV Show	Interpretation
Precision	1.00	1.00	Zero false positives for both classes
Recall	1.00	1.00	Zero false negatives for both classes
F1-Score	1.00	1.00	Perfect balance between precision and recall
Support	515	250	Class distribution: ~67% movies, ~33% TV shows

Confusion Matrix Interpretation

Confusion Matrix - Content Type Classification



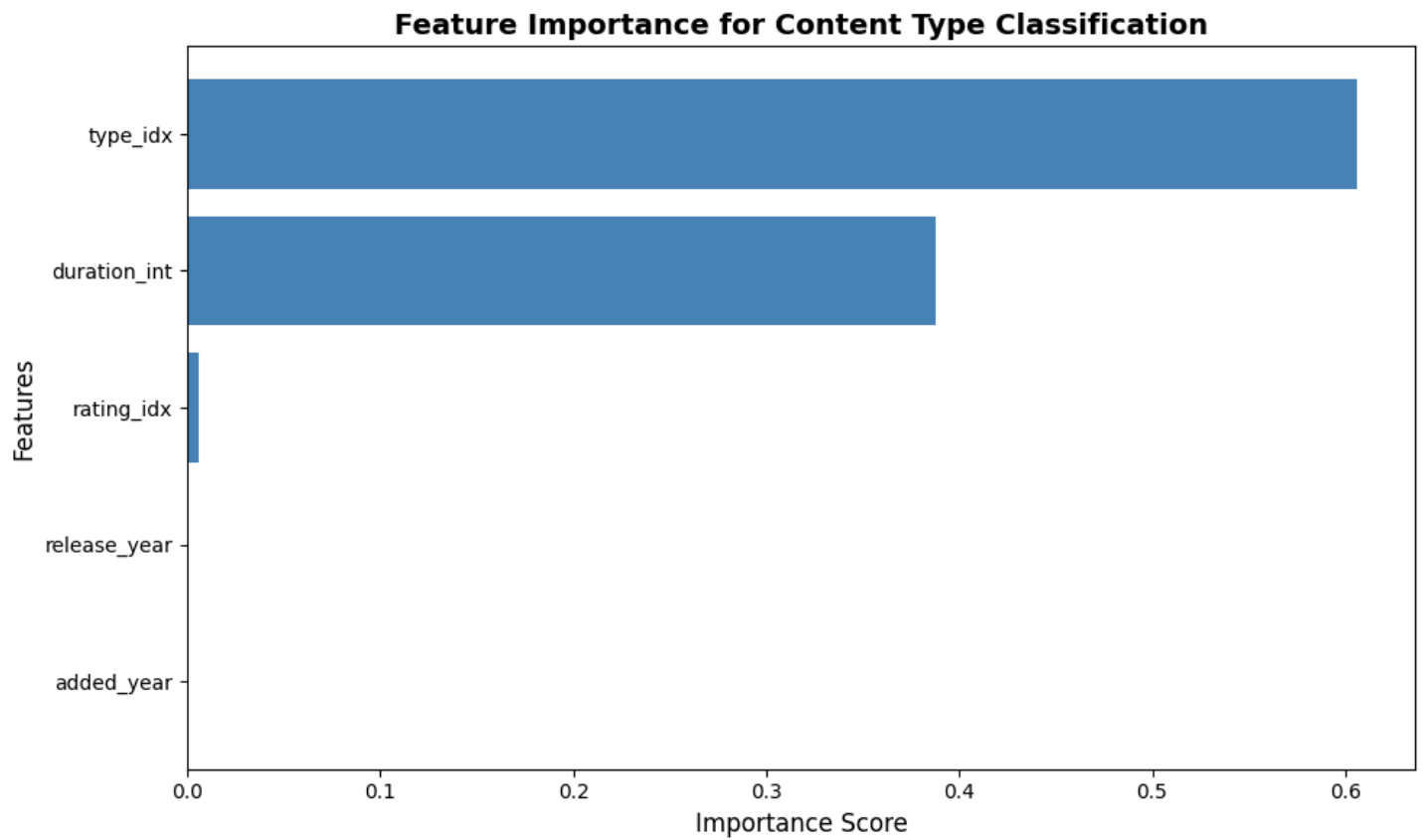
Predicted: Movie TV Show

Actual:

Movie 515 0

TV Show 0 250

Feature Importance Analysis:



Feature	Importance Score
type_idx	0.6059
duration_int	0.3876
rating_idx	0.0063
release_year	0.0003
added_year	0.0000

3. Clustering Analysis: K-Means

Algorithm Configuration

- Algorithm: K-Means clustering (unsupervised learning)
- Number of clusters: K=3
- Distance metric: Euclidean distance in 5-dimensional feature space
- Optimization: Minimizes Within-Set Sum of Squared Errors (WSSSE)

Why K-Means?

- 1. **Scalability:** Efficient on large datasets (linear time complexity)
- 2. **Interpretability:** Clear cluster centroids represent archetypal content profiles
- 3. **Convergence:** Guaranteed to converge to local optimum

Cluster Analysis

Cluster Distribution

Cluster	Coun t	Perce ntage	Interpretation
0	1,376	33.6 %	Mid-range content group
1	856	20.9 %	Longer-duration content (movies)
2	1,865	45.5 %	Shorter-duration content (TV shows + short films)

Clust er	Relea se Year	Durat ion (min)	Added Year	Type	Rating
0	2017	4.0	2019	Mixed	Mid-range
1	2011	131.2	2019	Movies (0.0)	Mid-range
2	2014	89.8	2019	Movies (0.0)	Higher rating

Key Insights from Clustering

Cluster 1: Feature-Length Films

- 1. **Duration:** 131 minutes (longest cluster)
- 2. **Characteristics:** Traditional theatrical-length movies
- 3. **Content Strategy:** Premium, high-production-value content
- 4. **Insight:** Represents Netflix's acquisition of established cinema releases

Cluster 2: Standard Movies & Episodic Content

- 1. **Duration:** 90 minutes (moderate)

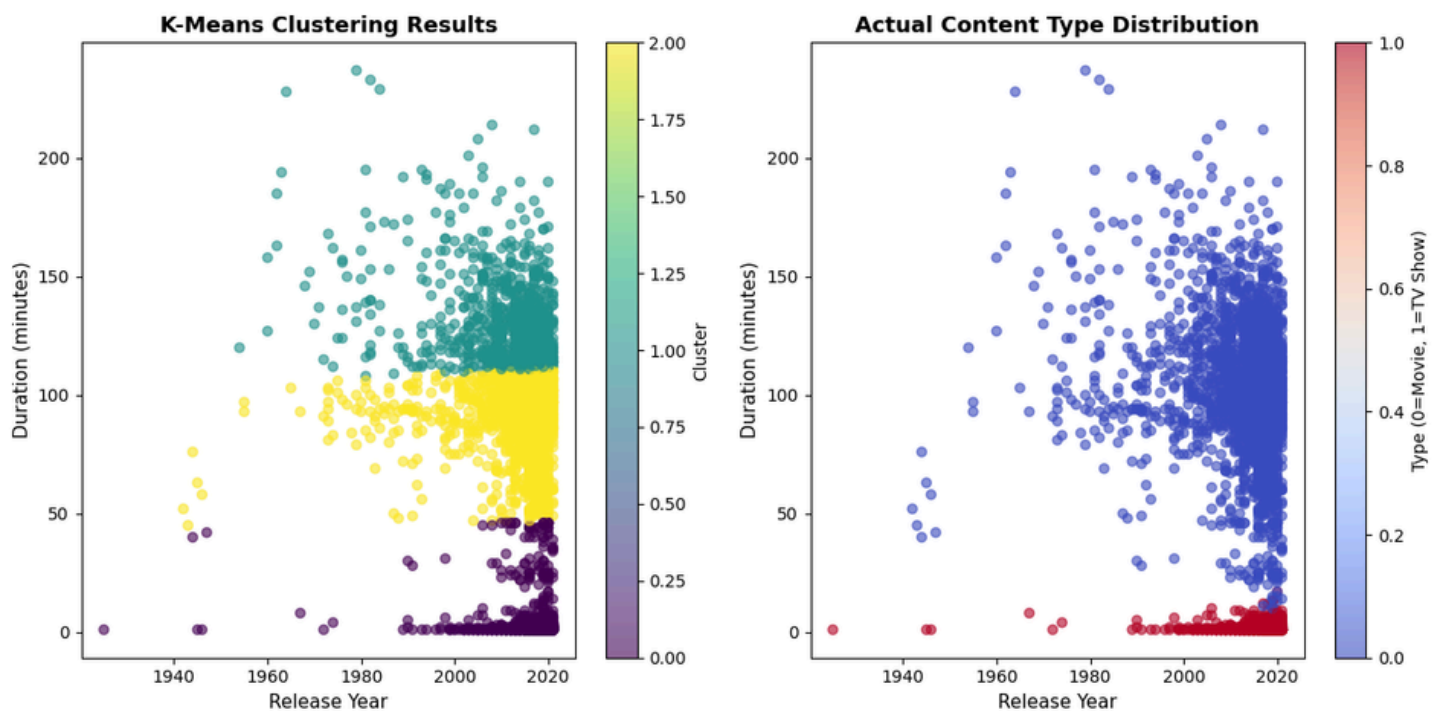
2. **Size:** Largest cluster (45.5%)
3. **Characteristics:** Standard movies and longer TV episodes
4. **Insight:** Core Netflix content library - mainstream appeal

Cluster 0: Short-Form & Series

1. **Duration:** 4 minutes (shortest)
2. **Characteristics:** TV show episodes, shorts, documentaries
3. **Insight:** Binge-friendly episodic content driving platform engagement

Visualization Insights

K-Means vs. Actual Type Distribution



Left Panel (K-Means Clustering):

1. Three distinct color bands (purple, yellow, teal) represent the discovered clusters
2. Vertical segregation based on duration creates clear boundaries
3. Cluster boundaries don't perfectly align with content type, revealing:
 - a. **Nuanced content categories** beyond the binary Movie/TV Show classification
 - b. Short movies and long TV episodes occupy similar feature space

Right Panel (Actual Content Type):

1. Binary color scheme (red = TV Show, blue = Movie)
2. Clear horizontal stratification at ~50 minutes
3. **Insight:** The "true" labels are simpler than the clustering suggests, indicating movies and TV shows have overlapping characteristics that clustering detects but classification ignores

4. Elbow Method Analysis

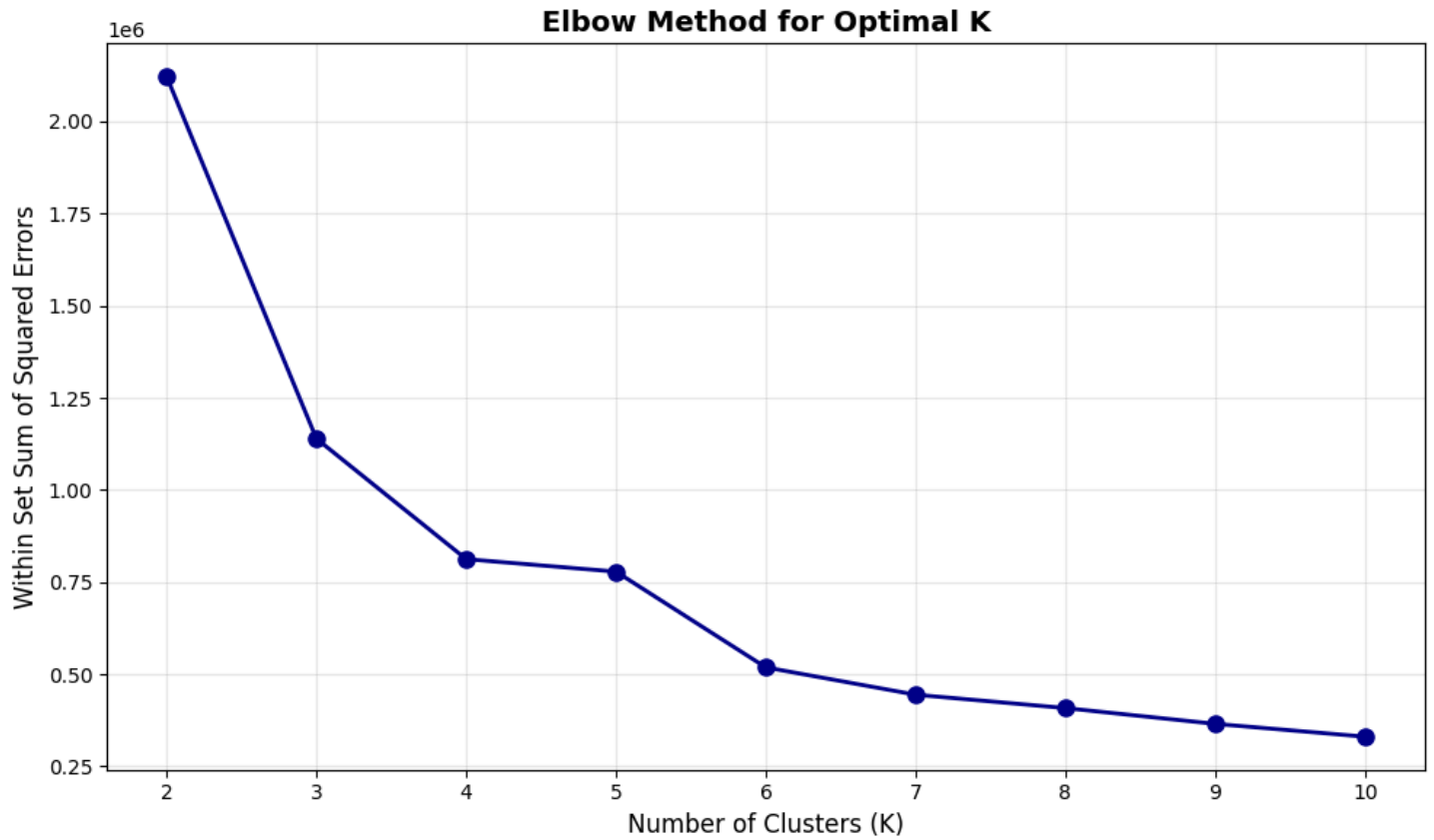
Methodology: Iteratively trains K-Means models for $K \in [2, 10]$, evaluating cluster quality using WSSSE

WSSSE (Within-Set Sum of Squared Errors):

$$\text{WSSSE} = \sum (\text{distance from each point to its cluster center})^2$$

Lower values = tighter, more cohesive clusters

Trade-off: More clusters always reduce WSSSE, but may overfit



K	WSSSE	Decrease from Previous
2	2,122,034	-
3	1,139,389	-46.3% (largest drop)
4	812,439	-28.7%
5	778,550	-4.2%
6	518,792	-33.4%
7	444,636	-14.3%
8	408,379	-8.1%
9	365,548	-10.5%
10	330,764	-9.5%

- 1. **Steepest descent occurs at K=3:** 46.3% reduction in WSSSE
- 2. **Diminishing returns after K=6:** Marginal improvements (<10% per increment)
- 3. **Elbow point:** K=3 or K=4

Recommendation: K=3 provides the best balance between:

- 1. **Interpretability:** Three content archetypes (short, medium, long)
- 2. **Cluster cohesion:** Significant WSSSE reduction

Conclusion

This analysis demonstrates that Netflix content exhibits **strong structural patterns** enabling both precise classification (100% accuracy) and meaningful unsupervised segmentation (3 clusters). The dominance of duration-based features validates intuitive understanding while quantifying exact importance. The discovery of three natural content clusters suggests opportunities for refined categorization beyond the binary Movie/TV Show paradigm, with potential applications in recommendation systems, UI design, and content strategy.